TRACK 1	Use Case Detail
AGENTS_001	**Adverse Media / Negative News Screening Copilot (Explainable)** Hackathon challenge: Find, summarize, and explain adverse media about an entity with traceable sources, entity disambiguation, and consistent risk decisioning. Key skills: Entity resolution, relevance scoring, explainable summarization, risk categorization Hints: Entity name → adverse media hits (with sources) → explainable risk score + reviewer workflow
AGENTS_002	**Agentic KYC Intelligence Platform** Hackathon challenge: Build a agentic KYC platform where multiple specialized agents collaborate to perform end-to-end customer due diligence—data extraction, enrichment, identity verification, compliance screening, financial profiling, and escalation—while maintaining explainability and human oversight. Key skills: Agentic orchestration, AMD ROCm development platform, data extraction, entity resolution, explainable risk scoring, human-in-the-loop design Hints: Customer onboarding request → AMD-orchestrated KYC agents (data & docs, enrichment, ID verification, screening, financial profile) → explainable KYC decision (approve / review / escalate) with evidence
AGENTS_003	**Policy & Document Comparison Assistant** Hackathon challenge: Build an AI tool that compares two policy documents (legacy vs modernized), highlights differences, and explains regulatory impact. Key skills: Text diff, semantic comparison Hints: Upload 2 PDFs → see comparison dashboard
AGENTS_004	**Customer complaint Classification & Routing Engine** Hackathon challenge: Automatically classify incoming complaints/service cases, prioritize them, and route to the right team with SLA awareness. Key skills: Text classification, prioritization logic Hints: Incoming case stream → live routing dashboard

TRACK 2	Use Case Detail
MULTIMODAL_001	**Drone-Based Cell Tower Inspection** A multi-modal pipeline that simultaneously ingests high-resolution 4K video feeds, thermal imaging, and geospatial telemetry from drones to automatically detect and classify structural anomalies in real-time.
MULTIMODAL_002	**Multi-modal Field Service Assistant** Multi-Modal AI powered assistant which can be used by Field service technicians through : computer vision (identify equipments or detect faults using mobile camera), Speech interface for hands-free interaction, provides document intelligence (technical manuals, SOPs, past cases) and agentic reasoning - step by step guidance and decision support
MULTIMODAL_003	**Real-time Sports Highlight Generator** Multi-modal pipeline analyzing video frames, commentary audio, and crowd signals to detect key events and auto-generate highlights, captions, and thumbnails.
MULTIMODAL_004	**Robotics & Industrial Autonomy (Physical AI)** Objective: Build AI that improves autonomy, safety, reliability, or productivity in industrial/robotic environments by turning sensor/vision/telemetry into fast decisions and actions. Example of what to build: A working prototype that ingests robot/plant data (telemetry, logs, images/video) and produces real-time guidance, anomaly alerts, or automated workflows (with clear KPIs like downtime reduction, defect reduction, safety compliance).

TRACK 3	Use Case Detail
FINETUNING_001	**Vulnerability-Aware Java Model** Hackathon challenge: Fine-tune a domain-specific LLM on labeled Java vulnerabilities and remediation examples to generate more accurate, consistent, and policy-aligned code fixes than prompt-only approaches. Key skills: Fine-tuning, dataset curation, model evaluation Hints: Vulnerability report → fine-tuned model → higher-precision Java fix (before vs. after comparison)
FINETUNING_002	**Telco specific Customer LLM** Generic public LLMs hallucinate when faced with proprietary telecom jargon, specific router hardware models, and complex internal billing codes. Fine-tuning an open-source model on years of anonymized telecom support transcripts and technical manuals to create a domain-expert support model that provides perfectly accurate, step-by-step troubleshooting.
FINETUNING_003	**Fraud Detection & Financial Crime Intelligence** Objective: Detect, prevent, and explain fraudulent activities by identifying abnormal patterns early, reducing false positives, and accelerating investigator decision making through AI driven insights. Example of what to build: An intelligent fraud detection prototype that ingests transactional, behavioral, and contextual data to deliver real time risk scoring, explainable alerts, and recommended next actions with human in the loop oversight.
FINETUNING_004	**Pathology image classifier:** Classify cancer subtype or simple histology pattern from images

AGENTS_005	**PPT / Executive Insight Generator** Hackathon challenge: Convert raw data or insights into executive-ready PowerPoint slides automatically. Key skills: Structured output, summarization Hints: Dataset → insights → auto-generated PPT
AGENTS_006	**AI-Driven Audit & Compliance Validator** Hackathon challenge: Create an AI agent that validates financial/insurance documents against compliance rules and produces an auditable report with confidence scores. Key skills: RAG, rule validation, explainability Hints: Compare a document vs rules → flag issues with evidence
AGENTS_007	**Telecom NOC (Network Operations Team) Agentic Copilot** NOC teams are overwhelmed with alarms, causing delayed root cause analysis and SLA breaches. A multi-agent system ingests alarms, logs, and KPIs, correlates events, identifies root cause, and suggests remediation actions and also executes API calls to reset routers or perform simple fixes. Integrates with ITSM to automate ticketing and resolution workflows.
AGENTS_008	**Autonomous Earnings Call Script & Presentation** A multi-agent workflow specifically designed for the Investor Relations function. Data Extraction Agent: Autonomously ingests raw quarterly financials and operational databases Predictive Analyst Agent: Analyzes past earnings transcripts, competitor performance, and current market sentiment to predict the most difficult questions institutional investors will ask. Drafting Agent: Synthesizes all inputs to autonomously generate a cohesive, compliant earnings call script, draft bullet points for the investor presentation deck, and create a "Q&A Cheat Sheet" for the CEO and CFO

MULTIMODAL_005	**Intelligent Image & Signal Processing** Objective: Derive actionable intelligence from high volume, high resolution visual or signal data using advanced AI techniques. Example of what to build: Models or applications that analyze images, video, or signal data to detect patterns, identify defects or anomalies, classify events, or enhance decision-making through visual or signal understanding
MULTIMODAL_006	**Value Chain Stage : Planning/Execution** **Autonomous Production Scheduling** Build a defect detection system with explainable GenAI outputs. Reduce false positives across variants.
MULTIMODAL_007	**Mutation to Mechanism to Therapy Reasoning** Build an AI driven system capable of reasoning from genomic mutation to biological mechanisms and potential therapeutic implications using biomedical data sources
MULTIMODAL_008	**Mutation to Mechanism to Therapy Reasoning:** Build an AI driven system capable of reasoning from genomic mutation to biological mechanisms and potential therapeutic implications using biomedical data sources

FINETUNING_005	**AI led Medical Review Assistant** - Assist medical reviewers to assess seriousness, suggest MedDRA codes, evaluate labelling status, and support causality assessment.

AGENTS_009	**Agentic Multi-modal AI-driven Content Moderation** AI agents that manage the entire moderation lifecycle.Perception Agent : Continuously ingests live video frames and audio transcripts to understand the interplay between visual and text modalities.• Policy Agent : When the Perception Agent flags an anomaly, it triggers the Policy Agent, which searches the platform's complex legal and community guideline databases to verify if a specific rule is broken.• Action Agent: Autonomously executes the takedown, shadowban, or age-restriction via API, and instantly generates a detailed "ban report" explaining exactly which policy was violated
AGENTS_010	**Intelligent Supply Chain & Operations** Objective: Increase agility, resilience, and efficiency across end to end operations through predictive and decision driven AI. Example of what to build: AI systems that forecast demand or supply, detect disruptions, optimize inventory or logistics, and recommend actions to balance cost, speed, and service levels.
AGENTS_011	**Next Generation Customer Experience (CX)** Objective: Deliver faster, more personalized, and more consistent customer interactions using intelligent, AI driven experiences. Example of what to build: AI solutions that assist agents or customers by understanding context, recommending next best actions, automating resolution steps, and summarizing interactions across channels.
AGENTS_012	**Chiplets & Advanced Packaging** Objective: Enable modular, heterogeneous systems by designing, validating, and optimizing chiplet-based architectures and advanced packaging approaches for performance, power, cost, and scalability. Example of what to build: A prototype workflow/tool that helps teams plan or validate multi-die architectures and packaging choices by integrating inputs like constraints, interface requirements, thermal/power limits, and system targets.
AGENTS_013	**Functional Safety (FuSa)** Objective: Accelerate safety-ready design and compliance-driven engineering by embedding functional safety considerations early and continuously across the engineering lifecycle. Example of what to build: A compliance-aware assistant/workflow that supports safety activities such as requirements traceability, hazard analysis support, evidence collection, review automation, and audit-ready reporting—with clear trace logs of decisions and artifacts.

MULTIMODAL_009	Computer vision driven understanding of product pricing and promotion fron shelf images and flyers

AGENTS_014	**AI Assisted Design & Verification – example AI led VLSI** Objective: Compress engineering cycles and boost productivity by using AI to assist with design, verification, and validation tasks—reducing repetitive effort and improving quality. Example of what to build: An AI-assisted engineering toolchain component that automates or accelerates one or more of: spec-to-RTL assistance, verification acceleration, test generation, bug triage, regression insights, or design-space exploration.
AGENTS_015	**Autonomous Operations – example Fab Autonomy** Objective: Improve yield, reliability, and resiliency through AI-driven autonomy—detecting drift early, optimizing schedules, reducing downtime, and enabling closed-loop operational intelligence. [Example of what to build: An autonomous operations prototype that combines telemetry + events + workflows to deliver: real-time anomaly detection, root-cause hints, optimization recommendations, and automated response orchestration with human oversight.
AGENTS_016	**Third Party & Vendor Risk Management** Objective: Reduce exposure to third party risks by proactively identifying financial, operational, and compliance threats across vendors and partners. Example of what to build: An AI solution that aggregates internal and external indicators to assess third party risk, detect early warning signals, and recommend mitigation actions.
AGENTS_017	**Value Chain Stage : Manufacturing Engineering** **Generative Process Optimization Advisor** Build an AI solution to recommend optimal process parameters under safety and operational constraints. Must ensure feasibility and compliance.
AGENTS_018	**Value Chain Stage : Procurement / Supply Chain** **Supplier Risk Intelligence** Predict supplier risks using financial, geopolitical, and ESG signals. Requires continuous learning from global events.
AGENTS_019	**Value Chain Stage : Operations/Maintenance** **Predictive Maintenance** Develop a self-learning agent that predicts failures and adapts models using real-time IoT data. Emphasis on edge deployment.

AGENTS_020	**Value Chain Stage : Product Development** **Product Design Validation** Create a GenAI assistant that validates designs against regulations and historical failures. Must merge rules-based and probabilistic logic.
AGENTS_021	**Value Chain Stage : Operations/Support** **Conversational Shopfloor Assistant** Develop a GenAI assistant for operator troubleshooting using enterprise knowledge. Must minimize hallucinations
AGENTS_022	**Value Chain Stage : Supply Chain / Inventory** **AI-Driven Inventory Optimization** Develop an agent that balances inventory levels against demand variability and supply disruptions. Must minimize both stockouts and excess inventory.
AGENTS_023	**Value Chain Stage : Aftermarket / Service** **Smart Warranty Analytics & Fraud Detection** Build a GenAI + ML solution to detect fraudulent warranty claims and identify systemic product issues. Requires pattern recognition across structured and text data.
AGENTS_024	**Value Chain Stage : Sales / Engineering** **Intelligent Product Configuration Assistant** Build a GenAI solution that helps configure complex products based on customer requirements and engineering constraints. Must ensure manufacturability.
AGENTS_025	**Value Chain Stage : Logistics / Distribution** **Agentic AI for Logistics Optimization** Create an autonomous agent that optimizes transportation routes, modes, and costs in real time. Must adapt to disruptions like delays or fuel price changes
AGENTS_026	**Autonomous Incident Diagnosis & Resolution Agent** Build an agentic AI system that continuously monitors infrastructure (logs, metrics, events), identifies anomalies, performs root cause analysis, and executes remediation actions (restart services, scale resources) with minimal human intervention.
AGENTS_027	**Self-Optimizing Workload Placement Agent** Develop an AI agent that dynamically places workloads across compute (VMs, containers, GPU/CPU nodes) based on cost, performance, and energy efficiency, learning from historical usage patterns.

AGENTS_028	**Autonomous Patch & Vulnerability Management Agent** Design an intelligent agent that scans systems for vulnerabilities, prioritizes patches based on risk/business impact, schedules deployments, validates success, and rolls back if needed.
AGENTS_029	**AI-driven Capacity Planning & Auto-Scaling Agent** Build an agent that forecasts future infrastructure demand using historical trends and automatically provisions/decommissions resources across clusters to avoid over/under-provisioning.
AGENTS_030	**Autonomous Change Impact Analysis Agent** Create an agent that evaluates planned infrastructure changes (patches, config updates, deployments), predicts potential impacts using historical incidents, and suggests safer rollout strategies.
AGENTS_031	**Intelligent Backup & Disaster Recovery Agent** Develop an AI agent that optimizes backup schedules, validates backup integrity, detects anomalies, and orchestrates autonomous disaster recovery with minimal RTO/RPO
AGENTS_032	**Unified Observability & RCA Agent (Cross-Tower)** Create an agent that correlates data across multiple towers (compute, storage, network, applications), generates human-readable RCA, and continuously learns from incidents to improve accuracy.
AGENTS_033	**Clinicial trial protocol risk predictor:** Predict whether a trial design may face recruitment, retention, or complexity issues
AGENTS_034	**Research data assistant (talk to data):** query and interpret heterogeneous datasets
AGENTS_035	**Smart Survellience** - AI driven segregation of cases (as Insufficient information, Case Confounded and Potential Index case) by referring narrative
AGENTS_036	AI in Personalization – Product recommendations and personalized promotions
AGENTS_037	Dynamic Content optimization basis the weather, geo-location, channel and personal characteristics
AGENTS_038	Dynamic UI for commerce
AGENTS_039	Social Media insights and triggers – trend prediction
AGENTS_040	Contact center reimaination with AI
AGENTS_041	Conversational commerce